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Individual Differences in Working Memory: Introduction to the Special Section

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This special section includes a set of 5 articles that examine the nature of inter- and intraindividual differences in working memory, using working memory span tasks as the main research tools. These span tasks are different from traditional short-term memory spans (e.g., digit or word span) in that they require participants to maintain some target memory items (e.g., words) while simultaneously performing some other tasks (e.g., reading sentences). In this introduction, a brief discussion of these working memory span tasks and their characteristics is provided first. This is followed by an overview of 2 major theoretical issues that are addressed by the subsequent articles—(a) the factors influencing the inter- and intraindividual differences in working memory performance and (b) the domain generality versus domain specificity of working memory—and also of some important issues that must be kept in mind when readers try to evaluate the claims regarding these 2 theoretical issues.

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Working memory has generated active, interdisciplinary research for the past quarter century. This special section of the *Journal of Experimental Psychology: General* focuses on one specific aspect of working memory research that has had a particularly strong, crosscutting impact on different disciplines within psychology—the nature of individual differences in working memory, broadly construed to include age-related variations (cognitive development and aging) as well as intraindividual variations due to various transient factors (e.g., the effect of anxiety). This special section presents a set of five articles that attempt to specify the nature and sources of individual differences in working memory and to demonstrate the usefulness of the individual differences approach in examining a wide range of theoretical and applied issues in psychology.

Although the articles vary widely in the target populations they deal with (e.g., young adults, children, elderly adults) as well as in the task domains (e.g., reading, numerical processing), one common thread that unites all the articles in this special section is that the studies used complex working memory span tasks (such as reading span, operation span, and counting span tasks) as research tools to assess individual differences in working memory. In this introduction, I first provide a brief overview of working memory span tasks and their characteristics, followed by a discussion of two major theoretical issues addressed by the subsequent articles.

Working Memory Span Tasks and Their Characteristics

Working memory span tasks require participants to simultaneously juggle both the storage and processing requirements that they impose. For example, in the case of the prevalently used reading span task (originally developed by Daneman & Carpenter, 1980), participants are required to read sentences aloud or sometimes verify the truthfulness of those sentences (the processing requirement) while trying to remember the last word of each sentence or some other target words for later recall (the storage requirement). For example, at the two-sentence level (Level 2), participants may be asked to read aloud the following sentences:

> Due to his gross inadequacies, his position as director was terminated abruptly.
>
> It is possible, of course, that life did not arise on the earth at all.

At the end of this trial, participants are expected to recall the two sentence-final words, *abruptly* and *all.* Other working memory span tasks are similar to this reading span task in that they require participants to maintain target items (e.g., words, digits, spatial orientations) while performing concurrent cognitive processing, such as simple arithmetic (Turner & Engle, 1989), counting (Case, Kurland, & Goldberg, 1982), or mental rotation (Shah & Miyake, 1996). Note that these working memory span tasks are different from traditional short-term memory tasks, such as digit span or word span, in that they essentially impose dual-task demands on participants (i.e., simultaneous processing and storage) and, hence, require more than a passive storage of target memory items.

These working memory span tasks have gained popularity as useful assessment tools in the field for a number of reasons. First, the demands that these tasks impose nicely matched the notion of working memory proposed by Baddeley and Hitch (1974) in terms of its emphasis on simultaneous processing and storage, which seem to characterize the demands typically imposed by complex cognitive tasks (e.g., language comprehension, mental arithmetic, reasoning, problem solving). Technically, the processing and storage requirements within a working memory span task do not exactly "simulate" the nature of simultaneous processing and storage required in a more naturalistic task in the sense that what one needs to process (e.g., reading random sentences) and store (e.g., arbitrarily chosen words) in a span task are often unrelated to each other, whereas they are closely related in everyday cognitive

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tasks (e.g., reading a newspaper article, mentally calculating the amount to tip). The addition of an explicit processing requirement, however, was considered to make working memory span tasks more than passive storage-only tasks and thereby make them more closely approximate the demands of complex cognitive tasks than do more traditional short-term memory span tasks.

Second and more important, working memory span tasks have been shown to be a better predictor of performance on complex cognitive tasks than their storage-oriented counterparts. For example, according to Daneman and Merikle's (1996) recent meta-analysis based on 77 previously published studies, the correlation between reading span and reading comprehension was .41 (with a 95% confidence interval [CI] of .38 to .44), which was significantly higher than the one between digit span and reading comprehension (.14 with a 95% CI of .10 to .18) or the one between word span and reading comprehension (.28 with a 95% CI of .23 to .33). In addition to reading comprehension, different versions of complex working memory span tasks have also been shown to be good predictors of various cognitive abilities, including spatial ability (Miyake, Friedman, Rettinger, Shah, & Hegarty, in press), reasoning ability (Kyllonen & Christal, 1990), general fluid intelligence (Engle, Tuholski, Laughlin, & Conway, 1999), and academic achievement (Gathercole & Pickering, 2000).

These characteristics have made working memory span tasks popular tools for examining the role of working memory in cognition and individual differences therein. They have been used not only to assess interindividual variations among normal young adults (Engle, Tuholski, et al., 1999; Kane, Bleckley, Conway, & Engle, 2001), but also to gain some insights into the nature of age-related changes in cognitive abilities among children (e.g., Gathercole & Pickering, 2000; Hitch, Towse, & Hutton, 2001) and elderly adults (e.g., Myerson, Hale, Rhee, & Jenkins, 1999; Lustig, Hasher, & May, 2001). They have also been used successfully among special populations, such as individuals with learning disabilities (Swanson & Ashbaker, 2000) and patients with aphasia (Caspari, Parkinson, LaPointe, & Katz, 1998). In addition, these span measures have been shown to be sensitive to various intra-individual differences and have been used to investigate the effects of various transient emotional or physiological states on working memory, such as mathematical anxiety (Ashcraft & Kirk, 2001) and smoking urges (Zwaan & Truitt, 1998). Working memory span tasks have also been used in some neuropharmacological studies that examine the effects of specific drugs (such as bromocriptine) on working memory (Kimberg, D'Esposito, & Farah, 1997).

Two of the studies included in this special section provide good illustrations of broad applicability and usefulness of working memory span tasks in addressing interesting applied questions. The Daneman and Hannon (2001) study, for example, takes advantage of the well-established link between working memory spans and verbal comprehension and reasoning ability to examine the validity of the reading comprehension portion of the Scholastic Assessment Test (SAT). The Ashcraft and Kirk (2001) study uses working memory span tasks to evaluate the on-line effects of mathematical anxiety on working memory performance in an attempt to specify the source(s) of poorer math performance that individuals with high levels of math anxiety often demonstrate.

Theoretical Issues Regarding Individual Differences in Working Memory

In addition to their wide use in addressing interesting applied questions, working memory span tasks have also been used to examine a number of important theoretical questions. The limited space does not allow me to provide a comprehensive review of all the major theoretical issues under debate (interested readers should refer to Caplan & Waters, 1999; MacDonald & Christiansen, in press; Miyake & Shah, 1999a). In this introduction, I focus instead on two central theoretical issues that have been debated in the literature and are examined in some depth in the articles in this special section. The two issues are (a) the factors that underlie individual or age-related differences in working memory and (b) the extent to which individual differences in working memory are domain specific or domain general.

Factors Underlying Individual and Age-Related Differences in Working Memory

The first issue, addressed primarily by the first three articles (Kane et al., 2001; Hitch et al., 2001; Lustig et al., 2001), concerns the nature of individual and age-related differences tapped by working memory span tasks. In other words, what do working memory span tasks really measure and what makes them better predictors of people's performance on complex cognitive tasks than more passive, storage-oriented span tasks? Although these are fundamental questions that surround working memory span research, there is currently no clear consensus among researchers (Miyake & Shah, 1999b; Saito & Miyake, 2000).

The original model proposed by Daneman and Carpenter (1980)—what Hitch et al. (2001) call the resource-sharing model—emphasized a trade-off between the processing and storage demands. According to this model, a working memory span task is a measure of a functional capacity of cognitive resources that can be flexibly allocated between the processing and storage activities within working memory. If an individual is skilled at, say, language processing, then fulfilling the concurrent processing requirement of the reading span task (i.e., reading aloud or sentence verification) consumes a relatively small amount of resources, enabling him or her to allocate a relatively large amount of the leftover resources for the maintenance of target memory items. In contrast, if a person is not skilled at language processing, then performing a concurrent language processing task consumes a lot of resources, leaving only a small amount available to support the storage of target words, hence, leading to a low span score. As this example illustrates, the resource-sharing model argues that performance on working memory span tasks is a reflection of the amount of resources one has available after the processing requirements of the tasks are met. Although there are a number of variants on this resource-sharing view (e.g., Just & Carpenter, 1992; Turner & Engle, 1989), they all incorporate the notion of trade-offs between processing and storage.

It is probably not an overstatement to say that this resource-sharing model has been the dominant view since its original conception and is still widely held. For example, the notion of processing-storage trade-off is adopted by Daneman and Hannon (2001) as well as by Ashcraft and Kirk (2001). In the mid- to late 1990s, however, a number of alternative proposals emerged that

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question or challenge the validity of the resource-sharing model. This special section focuses on three such proposals (for other theoretical proposals, see Ericsson & Kintsch, 1995; MacDonald & Christiansen, in press; Waters & Caplan, 1996).

A first alternative proposal represented here is Engle, Kane, and Tuholski's (1999) controlled attention view. This view proposes that working memory capacity, as measured by working memory span tasks, is determined jointly by short-term memory capacity and what they term "controlled attention" ability, which they characterize as a domain-general, limited attentional capacity for performing controlled processing or sustaining focus on task-relevant information in the face of interfering or distracting stimuli. In terms of Baddeley and Hitch's (1974) tripartite model of working memory, Engle, Kane, and Tuholski's proposal is analogous to the view that working memory capacity equals the capacity of the slave system (e.g., the phonological loop in the case of verbally oriented span tasks) plus the efficiency of central executive functioning. Note that this controlled attention view focuses on the role of executive control (i.e., controlled attention) and storage, rather than processing and storage, in working memory span performance and, hence, avoids the notion of processing-storage trade-off. In this special section, Kane et al. (2001) present two experiments that tested a prediction derived from the controlled attention view, namely that if a working memory span task strongly reflects one's controlled attention ability, then it should be able to reliably predict a person's performance on a relatively simple attention task (e.g., the so-called antisaccade task) that requires deliberate, controlled processing but does not place much demand on storage per se.

Another proposal that presents a clear alternative to the resource-sharing model is the task-switching hypothesis (Towse, Hitch, & Hutton, 1998, 2000). According to this view, working memory span tasks require participants to switch back and forth between the processing and storage requirements. The key aspect of this hypothesis is that instead of postulating a direct, trading relationship between the processing and storage requirements, it assumes an indirect relationship between the two; specifically, the efficiency with which one can perform the processing component of a span task affects the duration for which the target items need to be retained in memory, which in turn determines the degree to which those target items are susceptible to forgetting. For this reason, the task-switching hypothesis can explain why an individual who is less efficient in processing tends to show poorer performance on working memory span tasks, without resorting to the trade-off notion. In this special section, Hitch et al. (2001) present the data from a longitudinal study of working memory performance among children that helps illuminate the strengths and weaknesses of the task-switching hypothesis and the resource-sharing hypothesis.

Another proposal that can be considered an alternative to the original resource-sharing hypothesis is an inhibition-based account put forth by Hasher and her colleagues (e.g., Hasher & Zacks, 1988). According to this account, the notion of the limited supply of resources, which is an integral part of the resource-sharing model, may be an unnecessary assumption. Instead of the sheer capacity or the sheer amount of resources one has available for allocation to processing and storage requirements, performance on working memory span tasks (or other complex cognitive tasks) may alternatively be constrained by one's ability to inhibit irrelevant information. If an individual is equipped with a good inhibitory mechanism, then he or she can restrict the content of working memory to only those pieces of information that are directly relevant to the current task goal. According to this view, then, the degree to which working memory is "clutter free" (or how well one can manage the contents of working memory) is what determines efficient cognitive performance, not the amount of working memory resources that one has available. Lustig et al. (2001) adopted this inhibition-based view and tested one of its predictions in a study involving both young and elderly adults. Specifically, they provide evidence suggesting that performance on working memory span tasks reflects, at least in part, one's ability to counter the effects of proactive interference caused by the memory items that one encountered on earlier trials of the span task (May, Hasher, & Kane, 1999).

Thus, the first three articles in this special section each raise a theoretical proposal that differs considerably from the original resource-sharing model and present evidence consistent with that alternative proposal. Although the proposal and the supporting evidence outlined in each article are intriguing, there are a number of issues that one needs to keep in mind in evaluating them.

One issue is that the factors that influence span scores themselves may not necessarily be the same as those that influence the magnitude of correlations between these span scores and measures of other complex cognitive tasks (Miyake & Shah, 1999b). Evidence presented in two articles in this special section (Hitch et al., 2001; Lustig et al., 2001) are particularly relevant here. Lustig et al. (2001), for example, demonstrate that the experimental manipulations considered to influence the degree of proactive interference within a task (e.g., an insertion of irrelevant tasks between trials, a change in the order of administering the trials) can indeed influence the span scores. As Lustig et al. themselves admit, however, it is not completely clear whether the presumed effect of proactive interference also underlies the age-related group differences found between the young and older adults as well as the interindividual differences within a single age group. A similar issue can also be raised with respect to the Hitch et al. (2001) results. The first part of the Hitch et al. study presents an intriguing finding that working memory span is significantly higher when the processing requirement that takes a long time to complete (e.g., a longer sentence to read) comes at the beginning of a trial (e.g., long–short–short for a Level 3 trial) than at the end of a trial (e.g., short–short–long). This is presumably because having to perform time-consuming processing at the end of a trial prolongs the duration for which one must maintain the target items in memory, thereby making them more susceptible to forgetting. Although this experimental effect suggests that the processing duration may be a factor that can influence one's span scores, it does not necessarily mean that this factor is also responsible for interindividual differences in span scores, because the effect of time-based decay may or may not be constant across individuals. Thus, it is important to conceptually separate the factors that can influence the span scores themselves from those that serve as the source(s) of individual or group (age-related) differences in working memory span performance.

Another important issue that needs to be kept in mind is that each proposal discussed here only deals with one specific aspect of individual differences in working memory and may not necessarily be mutually exclusive (even in relation to the resource-sharing

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model; see Hitch et al., 2001, for more detailed discussions). Given the complexity of working memory span tasks, it is quite likely that multiple factors contribute to constrain one's span scores, with some factors affecting them equally across individuals or across different age groups and with some other factors being more directly responsible for individual or group differences. Thus, each theoretical proposal should be regarded as providing an important piece of the puzzle that eventually needs to be put together to more fully understand and specify the nature of inter- and intraindividual differences in working memory and complex cognition (Saito & Miyake, 2000).

Domain Generality Versus Domain Specificity of Working Memory

The second major theoretical issue addressed in this special section concerns the domain generality versus domain specificity of working memory. Each article adopts a certain perspective on this issue, and some of them (Ashcraft & Kirk, 2001; Daneman & Hannon, 2001; Hitch et al., 2001) also present relevant data in the verbal and numerical domains.

In the original article, Daneman and Carpenter (1980) presented a domain-specific view, based on a single resource pool notion. Consider, for example, an individual who, for whatever reason, is highly efficient in reading but not in arithmetic processing. If this individual is performing the reading span task, then he or she needs to allocate only a small amount of resources to the processing component (i.e., reading), so a large amount of resources are available for the maintenance of target words. The same person, however, should have more difficulty performing the operation span task because a large amount of resources needs to be allocated for the processing requirement (i.e., arithmetic), thereby leaving only a small amount of resources available for the storage of target memory items.

Although it makes intuitive sense in the context of the resource-sharing model, this domain-specific view has been highly controversial in individual differences studies of working memory. Some researchers have argued for a domain-specific view (Daneman & Tardif, 1987; Shah & Miyake, 1996), although the extent of separability assumed within working memory is somewhat different from proposal to proposal. For example, using the reading span task and the spatial span task (in which participants are required to keep track of spatial orientations while performing spatial transformations), Shah and Miyake (1996) demonstrated that the reading span correlates well with reading comprehension measures but not with spatial ability measures, whereas the spatial span shows the opposite pattern (see also Friedman & Miyake, 2000). Although they did not deny the existence of a domain-general factor, they argued for the necessity to recognize domain-specific components within working memory, at least verbal–phonological and visuospatial. In contrast, other researchers have argued for a domain-general view, emphasizing the unitary nature of working memory (Turner & Engle, 1989; Kyllonen, 1993). Turner and Engle (1989), for example, demonstrated that not only the reading span task but also the operation span task (which involves numerical rather than verbal processing) is a significant predictor of reading ability, thereby demonstrating some degree of domain generality.

These seemingly contradictory claims appear to be more apparent than real, however. Although the issue has not been completely resolved, there now seems to be some general consensus regarding this domain specificity versus domain generality debate, namely that working memory span tasks are highly complex and, hence, likely include both domain-specific and domain-general components (Engle, Kane, et al., 1999; Miyake & Shah, 1999b). The data presented in this special section are consistent with this view and again point to the multifaceted nature of working memory span tasks.

The evidence of some domain-general components is provided in a number of studies in this special section. Kane et al. (2001), for example, argued for the domain-general nature of working memory (or, more precisely, the domain-general nature of controlled attention) and provided support for this view by demonstrating that the verbal–numerical working memory span task (the operation span) can significantly predict people's performance on an oculomotor–spatial task (i.e., the antisaccade task). In addition, the Hitch et al. (2001) and the Daneman and Hannon (2001) studies demonstrate that verbally based and numerically based span tasks both correlate with measures of verbal ability (e.g., word reading in children and comprehension in adults, respectively).

At the same time, however, there is also some evidence of domain specificity in the reported data. For example, despite signs of some domain generality, both Hitch et al. (2001) and Daneman and Hannon (2001) showed that verbally based and numerically based span tasks show somewhat different patterns of correlations with performance on other cognitive tasks. Daneman and Hannon, for example, have shown that the reading span task tends to be a better predictor of reading comprehension ability than the operation span task, even though both tasks significantly predict reading comprehension (see also Daneman & Merikle, 1996). Furthermore, Ashcraft and Kirk (2001) have shown that math anxiety has clear on-line effects on working memory performance that tend to be more pronounced for numerical tasks than for verbal tasks. These results suggest that it is important to consider not just the domain-general but also domain-specific aspects of working memory.

Acknowledging this dual nature of working memory, Engle, Kane, et al. (1999) recently argued for a hierarchical view that recognizes both its domain-general and domain-specific aspects:

> Just as in intelligence research, general working memory factors appear to account for too much variance [in individual differences in cognition] to be ignored. However, in some studies (e.g., Shah & Miyake, 1996), significant variance is left to be explained beyond that accounted for by a general factor. We suggest that the behavioral, neuropsychological, and neuroanatomical evidence supports such a hierarchical view of working memory/attention. The specific factors correspond primarily to the domain of to-be-stored information, but the general factor transcends the domain of processing (Engle, Kane, et al., 1999, p. 125; for a similar view, see Cornoldi & Vecchi, 2000).

Such a hierarchical view not only provides a convenient summary of findings that have been accumulated in the field but also serves to remind researchers of the importance of considering both domain-general and domain-specific effects and specifying the relative extent of their respective contributions to individual and group differences in working memory performance.

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In this regard, I would like to point out some issues that must be taken into consideration in specifying the relative degree of domain generality versus domain specificity on the basis of individual differences studies. One important issue is that there may be some systematic age-related or population differences in the relative extent to which the domain-general and domain-specific components contribute to working memory span performance. For example, one intriguing possibility, suggested by recent neuroimaging evidence using simpler, more storage-oriented tasks (Reuter-Lorenz et al., 2000), is that as people get older, they tend to show less differentiation of working memory resources and, consequently, older adults may show stronger domain-general patterns than younger adults.

Another important point is that there are various factors that can bias the result of a study in one direction or the other. Consider, for example, the factors that might bias the result toward domain specificity. One such factor is the low reliability of working memory span measures. Although these measures have shown a relatively high degree of internal reliability (e.g., split-half reliability or Cronbach's alpha), the test–retest reliability may not always be high (Waters & Caplan, 1996; but see Klein & Fiss, 1999). Because the reliability of a measure puts a constraint on the magnitude of its correlations with other measures, the lack of zero-order correlations, which is often interpreted as evidence for separable systems or components, could possibly reflect the low reliability of the measures. Another factor that might bias the results toward domain specificity is the restrictions of range in the target population. It is well known that a restricted range can yield low intercorrelations of the measures (Legree, Pifer, & Grafton, 1996), but this point has not been explicitly acknowledged in many individual differences studies of working memory involving a cognitively advantaged group of college students. There are also some factors that can bias the results toward domain generality. Two such factors are interindividual differences in overall levels of motivation (i.e., some people are more motivated than others) and differences in the degree to which participants understand complex task instructions for the working memory span tasks (i.e., some people often fail to gain a complete understanding of what they are supposed to do for working memory span tasks). These factors are particularly relevant if the tasks are administered in group sessions or in a completely automated manner so that the experimenter cannot closely monitor all participants' behaviors or overall levels of understanding. These potential biases suggest that it is important to take these factors into consideration when one designs or evaluates an individual differences study that concerns the domain generality versus domain specificity issue.

Concluding Remarks

Since their original conception, working memory span tasks have served as useful tools to address interesting applied and theoretical questions. For these span tasks to remain useful in future research, however, it is necessary to carefully reexamine what abilities or cognitive mechanisms they really tap and how such abilities or mechanisms contribute to overall performance on the span tasks as well as to their correlations with measures of complex cognitive abilities. I hope that this special section provides an insightful step toward this important goal in working memory research.

References

Ashcraft, M. H., & Kirk, E. P. (2001). The relationships among working memory, math anxiety, and performance. *Journal of Experimental Psychology: General, 130,* 224–237.

Baddeley, A. D., & Hitch, G. J. (1974). Working memory. In G. H. Bower (Ed.), *The psychology of learning and motivation: Advances in research and theory* (Vol. 8, pp. 47–89). New York: Academic Press.

Caplan, D., & Waters, G. S. (1999). Verbal working memory and sentence comprehension. *Behavioral and Brain Sciences, 22,* 77–126.

Case, R., Kurland, M. D., & Goldberg, J. (1982). Operational efficiency and the growth of short-term memory span. *Journal of Experimental Child Psychology, 33,* 386–404.

Caspari, I., Parkinson, S. R., LaPointe, L. L., & Katz, R. C. (1998). Working memory and aphasia. *Brain and Cognition, 37,* 205–223.

Cornoldi, C., & Vecchi, T. (2000). Mental imagery in blind people: The role of passive and active visuospatial processes. In M. A. Heller (Ed.), *Touch, representation, and blindness* (pp. 143–181). New York: Oxford University Press.

Daneman, M., & Carpenter, P. A. (1980). Individual differences in working memory and reading. *Journal of Verbal Learning and Verbal Behavior, 19,* 450–466.

Daneman, M., & Hannon, B. (2001). Using working memory theory to investigate the construct validity of multiple-choice reading comprehension tests such as the SAT. *Journal of Experimental Psychology: General, 130,* 208–223.

Daneman, M., & Merikle, P. M. (1996). Working memory and language comprehension: A meta-analysis. *Psychonomic Bulletin & Review, 3,* 422–433.

Daneman, M., & Tardif, T. (1987). Working memory and reading skill reexamined. In M. Coltheart (Ed.), *Attention and performance XII: The psychology of reading* (pp. 491–508). Hillsdale, NJ: Erlbaum.

Engle, R. W., Kane, M. J., & Tuholski, S. W. (1999). Individual differences in working memory capacity and what they tell us about controlled attention, general fluid intelligence, and functions of the prefrontal cortex. In A. Miyake & P. Shah (Eds.), *Models of working memory: Mechanisms of active maintenance and executive control* (pp. 102–134). New York: Cambridge University Press.

Engle, R. W., Tuholski, S. W., Laughlin, J. E., & Conway, A. R. A. (1999). Working memory, short-term memory, and general fluid intelligence: A latent variable approach. *Journal of Experimental Psychology: General, 128,* 309–331.

Ericsson, K. A., & Kintsch, W. (1995). Long-term working memory. *Psychological Review, 102,* 211–245.

Friedman, N. P., & Miyake, A. (2000). Differential roles for visuospatial and verbal working memory in situation model construction. *Journal of Experimental Psychology: General, 129,* 61–83.

Gathercole, S. E., & Pickering, S. J. (2000). Working memory deficits in children with low achievements in the national curriculum at 7 years of age. *British Journal of Educational Psychology, 70,* 177–194.

Hasher, L., & Zacks, R. T. (1988). Working memory, comprehension, and aging: A review and a new view. In G. H. Bower (Ed.), *The psychology of learning and motivation* (Vol. 22, pp. 193–225). San Diego, CA: Academic Press.

Hitch, G. J., Towse, J. N., & Hutton, U. (2001). What limits children's working memory span? Theoretical accounts and applications for scholastic development. *Journal of Experimental Psychology: General, 130,* 184–198.

Just, M. A., & Carpenter, P. A. (1992). A capacity theory of comprehension: Individual differences in working memory. *Psychological Review, 99,* 122–149.

Kane, M. J., Bleckley, M. K., Conway, A. R. A., & Engle, R. W. (2001). A controlled-attention view of working-memory capacity. *Journal of Experimental Psychology: General, 130,* 169–183.

Kimberg, D. Y., D'Esposito, M., & Farah, M. J. (1997). Effects of

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bromocriptine on human subjects depend on working memory capacity. *NeuroReport, 8,* 3581–3585.

Klein, K., & Fiss, W. H. (1999). The reliability and stability of the Turner and Engle working memory task. *Behavior Research Methods, Instruments, & Computers, 31,* 429–432.

Kyllonen, P. C. (1993). Aptitude testing based on information processing: A test of the four-sources model. *Journal of General Psychology, 20,* 375–405.

Kyllonen, P. C., & Christal, R. E. (1990). Reasoning ability is (little more than) working-memory capacity?! *Intelligence, 14,* 389–433.

Legree, P. J., Pifer, M. E., & Grafton, F. C. (1996). Correlations among cognitive abilities are lower for higher ability groups. *Intelligence, 23,* 45–57.

Lustig, C., May, C. P., & Hasher, L. (2001). Working memory span and the role of proactive interference. *Journal of Experimental Psychology: General, 130,* 199–207.

MacDonald, M. C., & Christiansen, M. H. (in press). Reassessing working memory: A comment on Just & Carpenter and Waters & Caplan. *Psychological Review.*

May, C. P., Hasher, L., & Kane, M. J. (1999). The role of interference in memory span. *Memory & Cognition, 27,* 759–767.

Miyake, A., Friedman, N. P., Rettinger, D. A., Shah, P., & Hegarty, M. (in press). How are visuospatial working memory, executive functioning, and spatial abilities related? A latent variable analysis. *Journal of Experimental Psychology: General.*

Miyake, A., & Shah, P. (1999a). *Models of working memory: Mechanisms of active maintenance and executive control.* New York: Cambridge University Press.

Miyake, A., & Shah, P. (1999b). Toward unified theories of working memory: Emerging general consensus, unresolved theoretical issues, and future research directions. In A. Miyake & P. Shah (Eds.), *Models of working memory: Mechanisms of active maintenance and executive control* (pp. 442–481). New York: Cambridge University Press.

Myerson, J., Hale, S., Rhee, S. H., & Jenkins, L. (1999). Selective interference with verbal and spatial working memory in young and older adults. *Journal of Gerontology: Psychological Sciences, 54B,* P161–P164.

Reuter-Lorenz, P. A., Jonides, J., Smith, E. E., Hartley, A., Miller, A., Marshuetz, C., & Koeppe, R. A. (2000). Age differences in frontal lateralization of verbal and spatial working memory revealed by PET. *Journal of Cognitive Neuroscience, 12,* 174–187.

Saito, S., & Miyake, A. (2000). A systematic evaluation of six hypotheses regarding the reading span test. *Japanese Psychological Review, 43,* 387–410.

Shah, P., & Miyake, A. (1996). The separability of working memory resources for spatial thinking and language processing: An individual differences approach. *Journal of Experimental Psychology: General, 125,* 4–27.

Swanson, H. L., & Ashbaker, M. H. (2000). Working memory, short-term memory, speech rate, word recognition and reading comprehension in learning disabled readers: Does the executive system have a role? *Intelligence, 28,* 1–30.

Towse, J. N., Hitch, G. J., & Hutton, U. (1998). A reevaluation of working memory capacity in children. *Journal of Memory and Language, 39,* 195–217.

Towse, J. N., Hitch, G. J., & Hutton, U. (2000). On the interpretation of working memory span in adults. *Memory & Cognition, 28,* 341–348.

Turner, M., & Engle, R. W. (1989). Is working memory capacity task dependent? *Journal of Memory and Language, 28,* 127–154.

Waters, G. S., & Caplan, D. (1996). The measurement of verbal working memory capacity and its relation to reading comprehension. *Quarterly Journal of Experimental Psychology, 49A,* 51–79.

Zwaan, R. A., & Truitt, T. P. (1998). Smoking urges affect language processing. *Experimental and Clinical Psychopharmacology, 6,* 325–330.

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